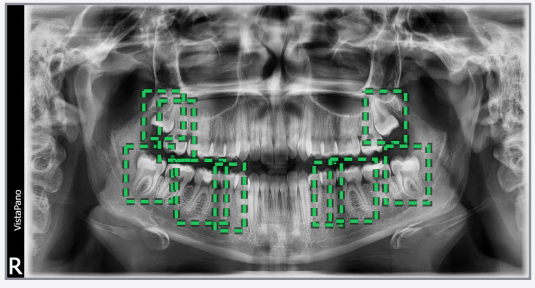
Two sources in use

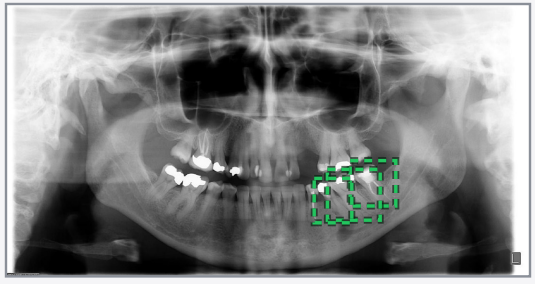
DENTEX — 2,332 images



quadrant, enumeration, diagnosis; 4 classes
only panoramic set with position + all 4 classes

- loader complete

Tufts Dental Database — 1,000 images



per-tooth boxes, expert pathology layer
periapical maps to DENTEX; no caries layer

- loader partial, 1 open question

Grounding pool and detector

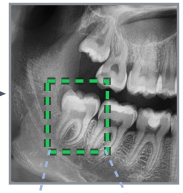
DENTEX
1,339 images
21,624 boxes

Tufts
1,000 images
25,184 boxes

2,339 images
46,808 boxes

YOLOv8m
32 FDI classes
5-fold CV

locate_tooth





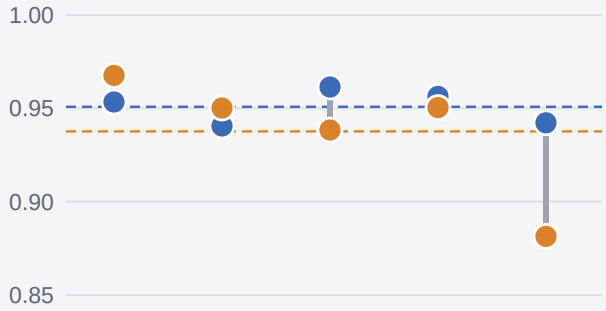
FDI 48: box, crop

annotated tooth: exact box
otherwise: detector box

trained

In-fold CV, target mAP50

- DENTEX-only, mean 0.9508
- DENTEX + Tufts, mean 0.9376



	fold 0	fold 1	fold 2	fold 3	fold 4
DENTEX-only	0.953	0.941	0.961	0.956	0.942
DENTEX + Tufts	0.968	0.950	0.938	0.950	0.881

Co-training with Tufts

raw val mAP50
- 0.5820
- 0.8695

precision
- 0.9637
- 0.9680

protocol differs from the CV

Held-out val



0.9593 mAP50
0.9864 precision

50 images, 46 pathological, 182 target boxes
best: DENTEX + Tufts, fold 1; deployed backend

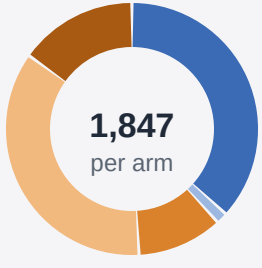
Eight candidate datasets, not used

- grounding-only 7 of 8
- diagnosis-labeled 1 of 8

Dental Radiography	1,272	no confirmed diagnosis taxonomy
DNS (UFBA-UESC)	543	FDI boxes in COCO, on request (follow-up)
Segmented Mandibles	232	mandible-only masks, not per tooth
TL-pano	197	FDI + quadrant labels, NC license (follow-up)
TK Tooth Number Code	188	numbering only, no diagnosis
Tunisia Panoramic	180	in progress, 1 open FDI question
Panoramic-Paraguay	135	thin documentation in the review
Panoramic-Caries-Seg	75	binary caries masks, pixel level

promotion rule: annotation semantics verified end-to-end, per dataset

Corpus cohorts



DENTEX 705
- 678 pathology
- 27 healthy

Tufts 1,142
- 202 overlap
- 660 healthy
- 280 all diseases